Geometrical Instruments in Practice

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

6 lecture sections · 10 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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2. Position angle convention
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1. Practical geometry on a chart

A double star is a clean laboratory for angular measurement. The two quantities are separation ρ and position angle P. Separation is the angular distance between components; position angle is measured from north through east. A ruler gives millimetres on a chart; the plate scale converts millimetres to arcseconds.

$$\rho = k\,L, \tag{1}$$

where k is the plate scale in arcsec/mm and L is the measured length in mm.

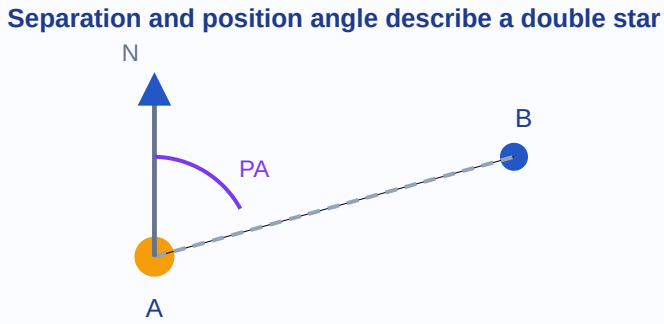


Figure 1. Plate scale converts chart millimetres into arcseconds.

2. Position angle convention

North is 0°, east 90°, south 180°, west 270°. If the companion is at position angle P, its sky components relative to the primary are

$$\Delta\alpha\cos\delta = \rho\sin P, \qquad \Delta\delta = \rho\cos P. \tag{2}$$

A protractor must be aligned with the chart’s north direction. If the chart is mirrored or rotated, the position angle must be corrected. Always state the convention and epoch.

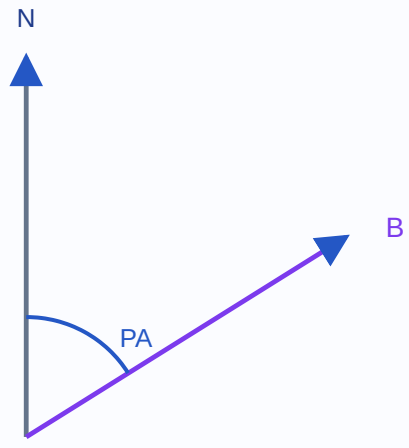


Figure 2. State the convention and the epoch when reporting measurements.

3. Plate scale and calibration

The plate scale can be calculated from focal length and pixel size, or measured from a known double star or star field. If a known separation ρ_0 measures L_0 mm, then $k = \rho_0 / L_0$. Calibration uncertainty propagates directly into every measurement.

Plate scale k is measured in arcsec/mm. A smaller k means more millimetres per arcsecond, so the same ruler error produces a smaller angular error. But a very small k may make the field too small or the image too dim.

4. Reading uncertainty

If the ruler reading uncertainty is σ_L mm, the angular uncertainty is

$$\sigma_\rho = k\sigma_L. \tag{3}$$

Position-angle uncertainty depends on separation: for a small positional error $\sigma_\perp$ perpendicular to the line, $\sigma_P \approx \sigma_\perp / \rho$ in radians. Wide pairs are easier to measure in angle; close pairs are harder.

5. Measuring procedure

Centre the primary, align north, measure the separation several times, rotate the chart or protractor to check alignment, and record the epoch. If using a micrometer or reticle, calibrate its scale. If using an image, measure centroid positions rather than eyeballing the centres. Repeat measurements and report the mean and scatter.

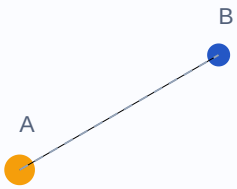
INTERACTIVE EXPERIMENT

Measure a double star on a chart.

The companion is placed at a known separation and position angle. A ruler reading error is converted into an angular uncertainty using the plate scale.

Ruler length	Measured separation	Angular uncertainty
8.00 mm	**8.00″**	**±0.50″**

Position angle
60°



Angular uncertainty = ruler uncertainty × plate scale. A smaller plate scale (arcsec/mm) reduces the angular effect of the same reading error.

Try this. Halve the plate scale k (arcsec/mm). What happens to the angular uncertainty for the same ruler error?

Snapshot of the experiment. Open the HTML edition to change the controls.

6. Reporting a double-star measurement

A complete report includes component identifiers, epoch, separation, position angle, plate scale, method, uncertainty and any resolution limit. If the pair is not resolved, state an upper limit rather than a false measurement. A good measurement is honest about what the instrument could and could not separate.

A reliable method. Calibrate the scale, align north, measure separation and angle independently, repeat, and propagate the reading error. State the epoch and convention.

7. Lecture + Problems: predict, calculate, check

Ten original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Separation from plate scale. A chart has plate scale 1″/mm. A double star measures 8 mm. Find the angular separation.

Worked solution →

[1] Problem 2 — Position angle. A companion lies due east of the primary. What is its position angle? What if it lies due north?

Worked solution →

[2] Problem 3 — Components. A pair has separation 8″ and position angle 60°. Find $\Delta\alpha \cos\delta$ and $\Delta\delta$.

Worked solution →

[2] Problem 4 — Reading uncertainty. A ruler has uncertainty 0.5 mm and the plate scale is 2″/mm. Find the angular uncertainty. How does it change if the plate scale is 0.5″/mm?

Worked solution →

[2] Problem 5 — Calibrating scale. A known double star with separation 10″ measures 5 mm on a chart. Find the plate scale. A target measures 12 mm; what is its separation?

Worked solution →

[1] Problem 6 — Mirrored chart. A chart is mirrored east–west. A companion appears at position angle 60°. What is the true position angle?

Worked solution →

[2] Problem 7 — Position-angle uncertainty. A pair has separation 5″. The perpendicular positional uncertainty is 0.1″. Estimate the position-angle uncertainty in degrees.

Worked solution →

[2] Problem 8 — Resolution limit. A telescope has a practical resolution of 1.5″. Can a resolved-image ruler method measure a 0.8″ double star as two components? What should be reported instead?

Worked solution →

[3] Problem 9 — Epoch and motion. A double star’s position angle changes at an approximately constant 2°/yr over a short observed arc. What is the change over 10 years under that approximation? Why must the epoch be stated, and why might the rate not stay constant over a full orbit?

Worked solution →

[3] Problem 10 — Design a measurement. You have a printed chart, a ruler and a protractor. Describe how to measure separation and position angle, and list two ways to reduce uncertainty.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.